UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 8 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

Complete the following taxonomy of parallel processor architecture with appropriate example.

Processor Organization	Single Instruction Multiple Data Stream	(i)
		(ii)
	Multiple Instruction Multiple Data Stream	(iii)
		(iv)
		(v)

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QUESTION A2 **(5 marks)**

Question 1 is based on the following placard in the Computer History Museum.

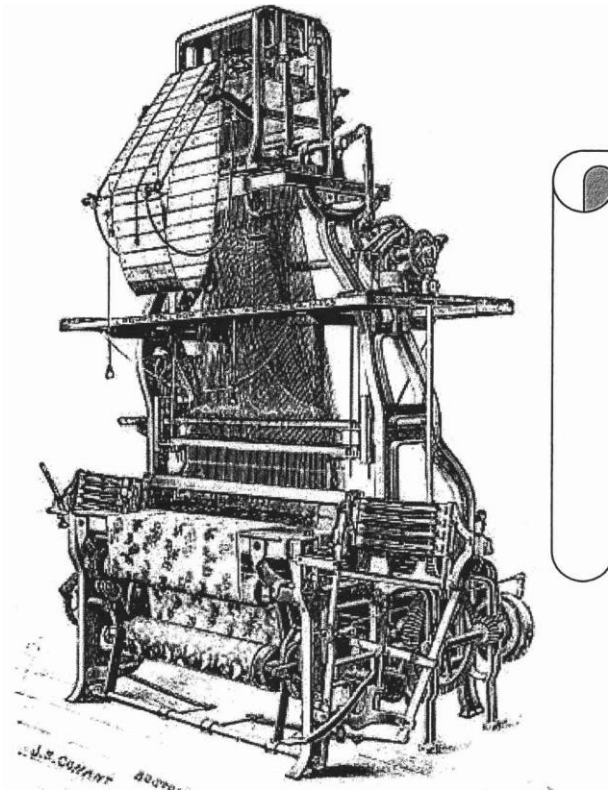


Figure 1: Machine X

The Jacquard System was developed by Joseph Marie Jacquard to automatically produce the intricate woven patterns. The deck of punched cards that controls the Machine X is located at the left center on the far side of the figure. In the early days of software piracy, the intricate fabric design card decks would be stolen by competing textile mills.

a) Name the Machine X in Figure 1.

b) In which computer generation was Machine X developed?

c) Name the possible computer component that has similar function as the deck of punch card in Machine X.

d) Identify the input and output for Machine X.

QUESTION A3 **(5 marks)**

Calculate -53 - 31 using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

QUESTION A4 **(5 marks)**

Given the following Figure 2.

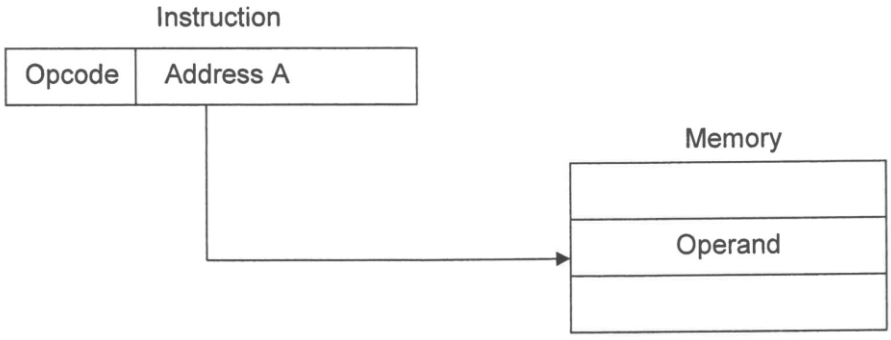


Figure 2

a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

c) List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in 9(a).

QUESTION A5 **(10 marks)**

a) Explain TWO (2) methods to handle multiple interrupts.

b) Give ONE (1) example of a non-maskable interrupt.

c) Why Direct Memory Access (DMA) is faster than interrupt-driven and programmed 1/0?

d) USB is known as a 'hot' connection. What does it mean by "hot connection"?

QUESTION A6 **(10 marks)**

a) Rank the following memory types based on price ranging from the cheapest to the most expensive. Cache, tape back-up, register, hard drive

b) Discuss ONE (1) benefit of cache memory.

c) Differentiate between write through and write back mechanisms in order to store data in cache.

QUESTION A7 **(10 marks)**

a) Discuss the whole operation of the machine cycle and its respective responsible computer components.

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b) Elaborate on what will happen in the machine cycle if the main memory is corrupted.

QUESTION A8 **(10 marks)**

Input output technique can be segregated into programmed, interrupt-driven and direct memory access (DMA).

a) What is DMA?

b) How can the DMA technique improve the overall computer performance?

c) Discuss TWO (2) major drawbacks of programmed and interrupt-driven input output technique that inspire hardware developer to introduce DMA.

d) Conventional access is depicted in Figure 2. Redraw Figure 2 to illustrate DMA architecture.

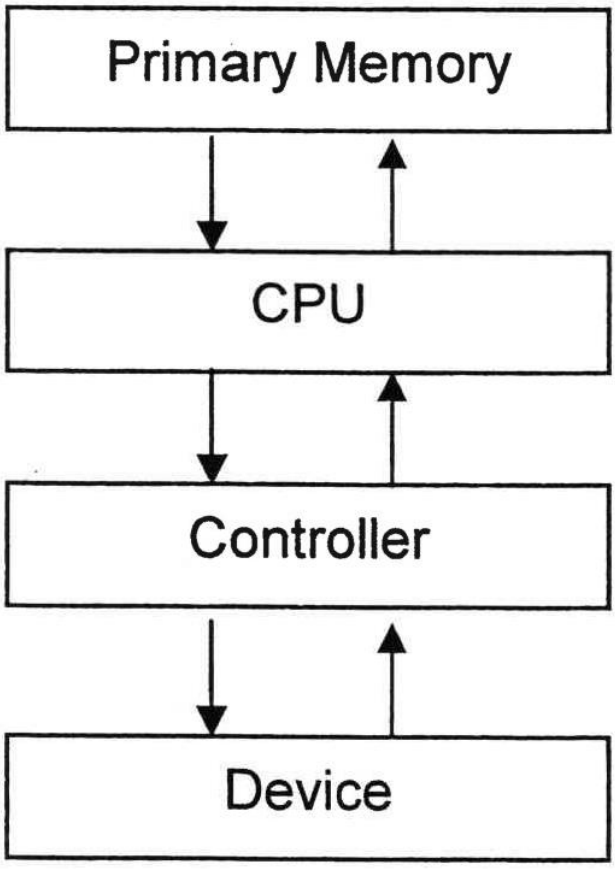


Figure 2: Conventional access architecture

QUESTION A9 **(10 marks)**

a) List TWO (2) buses types that are available in the computer system.

b) Differentiate TWO (2) characteristics of system interconnection and CPU interconnection.

c) Explain the different types of data transfer timing.

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QUESTION A10 **(10 marks)**

Answer both parts. Show your complete workflow.

a) Convert -31.5 to IEEE 754 single precision. Give the answer in hexadecimal.

b) Convert the IEEE 754 single-precision value 0xC1360000 to decimal.

PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		

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